

AI STUDYBUDDY API

PROJECT PLANNING PHASE

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PROJECT PLANNING OVERVIEW

1. Introduction

The **Project Planning Phase** is the initial stage of developing the AI StudyBuddy API. It provides a structured plan for designing, developing, testing, and deploying the API.

The AI StudyBuddy API acts as a communication layer between the student application, database, and Artificial Intelligence services. It provides different API endpoints for user authentication, AI chat, notes, quizzes, flashcards, study planning, and progress tracking.

The main purpose of planning is to clearly define the project objectives, API requirements, resources, development schedule, and possible risks before implementation begins.

1.1 Project Vision

The vision of the AI StudyBuddy API is to provide a reliable and scalable backend service that helps students access intelligent learning features through a simple API.

The API should allow applications to:

- Authenticate students securely.
- Send questions to the AI service.
- Generate and manage study notes.
- Generate and conduct quizzes.
- Create and manage flashcards.
- Manage study plans.
- Track student learning progress.
- Search for educational resources.

1.2 Project Objectives

The main objectives of the API project are:

1. To develop a RESTful API for AI StudyBuddy.

2. To provide secure user authentication.
3. To integrate AI-based question answering.
4. To provide APIs for creating and managing study materials.
5. To store student information and learning data.
6. To provide standardized JSON responses.
7. To validate API requests and handle errors.
8. To design an API that can be extended in the future.

1.3 Project Scope

Module	Planned API Functions
Authentication	Register, login, logout, profile
AI Chat	Ask questions and retrieve responses
Notes	Create, view, update, delete and generate notes
Quiz	Generate, retrieve and submit quizzes
Flashcards	Create, view and delete flashcards
Study Planner	Create and manage study activities
Progress	Store and retrieve learning progress
Resources	Search educational resources

API REQUIREMENT AND RESOURCE PLANNING

2. API Requirement Planning

The API requirements describe the functions that the backend must provide to the AI StudyBuddy application.

2.1 Authentication API

The authentication module manages student accounts.

Planned endpoints:

```
POST /api/auth/register
POST /api/auth/login
GET /api/auth/profile
PUT /api/auth/profile
POST /api/auth/logout
```

These endpoints will be responsible for account creation, login, profile management, and logout.

2.2 AI Chat API

The AI Chat module allows students to communicate with the AI assistant.

```
POST /api/chat
GET /api/chat
DELETE /api/chat/{id}
```

Example request:

```
{
  "message": "Explain the concept of inheritance in Java."
}
```

Example response:

```
{
  "success": true,
  "answer": "Inheritance is a feature of object-oriented programming..."
}
```

2.3 Notes API

The Notes API manages student study notes.

```
GET    /api/notes
POST   /api/notes
POST   /api/notes/generate
PUT    /api/notes/{id}
DELETE /api/notes/{id}
```

2.4 Quiz API

The Quiz API provides AI-generated quizzes and quiz submission.

```
POST /api/quizzes
GET  /api/quizzes/{id}
POST /api/quizzes/{id}/submit
```

2.5 Resource Planning

The project requires hardware, software, development tools, and human resources.

Resource	Purpose
Computer	API development
Code Editor	Writing backend code
Database	Storing application data
Postman	API testing
Git	Version control
Internet	AI and external API access
AI Service	AI-generated responses

2.6 Suggested Technology Stack

Component	Technology
API Architecture	REST
Data Format	JSON
Backend	Python / Node.js / PHP
Database	MySQL / PostgreSQL
Authentication	JWT / Session
API Testing	Postman
Version Control	Git
AI Integration	AI Model API

API DEVELOPMENT SCHEDULE AND TASK PLANNING

3. Development Planning

The development process is divided into smaller tasks so that each part of the API can be developed and tested systematically.

3.1 Project Development Schedule

Phase	Activity	Duration
1	Requirement Analysis	1 Week
2	Project Planning	1 Week
3	API Architecture Design	1 Week
4	Database Design	1 Week
5	Authentication API	1 Week
6	AI Chat API	1 Week
7	Notes and Quiz APIs	2 Weeks
8	Flashcard and Progress APIs	1 Week
9	API Testing	1 Week
10	Deployment and Documentation	1 Week

3.2 API Development Tasks

The following tasks are planned for development:

Task 1 – Project Setup

- Create backend project.
- Configure development environment.
- Configure database connection.
- Create API folder structure.

Task 2 – Authentication

- Create user registration.
- Implement login.
- Implement authentication tokens.
- Create profile management.
- Implement logout.

Task 3 – AI Integration

- Connect the AI service.
- Create AI chat endpoint.
- Validate student questions.
- Process AI responses.
- Store chat history.

Task 4 – Study Material APIs

Develop APIs for:

- Notes.
- Quizzes.
- Flashcards.
- Study plans.
- Learning resources.

Task 5 – Progress Tracking

The API will record activities such as quiz scores and completed study tasks.

Example:

```
{
  "user_id": 101,
  "activity": "Python Quiz",
  "score": 85,
  "completed_at": "2026-09-25"
}
```

3.3 API Request Flow

```
Student Application
    ↓
HTTP Request
    ↓
AI StudyBuddy REST API
    ↓
Authentication
    ↓
Request Validation
    ↓
Business Logic
    ↓
Database / AI Service
    ↓
JSON Response
    ↓
Student Application
```

3.4 API Standards

The project will follow standard HTTP methods:

Method	Purpose
GET	Retrieve data
POST	Create or process data
PUT	Update data
DELETE	Remove data

The API will use JSON for requests and responses.

RISK, SECURITY, TESTING AND MONITORING PLAN

4. Risk Planning

Risk planning identifies possible problems that could affect the API project.

Risk	Impact	Planned Solution
AI service unavailable	AI features stop working	Error handling and retry mechanism
Database failure	Data may become unavailable	Regular backups
Unauthorized access	Security problem	Authentication and authorization
Invalid input	API errors	Request validation
High API requests	Performance problems	Rate limiting
Development delay	Late project completion	Task monitoring
Incorrect AI response	Learning information may be unreliable	Response validation and user verification

4.1 Security Planning

Security is an important part of the AI StudyBuddy API.

The following security measures are planned:

- HTTPS for secure communication.
- Password hashing.
- Authentication tokens.
- Authorization checks.
- Input validation.
- API rate limiting.
- Protection of AI service credentials.
- Secure database access.
- Standardized error responses.
- Prevention of unauthorized data access.

4.2 API Testing Plan

Each endpoint will be tested before deployment.

Test	Expected Result
Valid login	Login successful
Invalid login	Error returned
Valid AI question	AI response returned
Invalid request	Validation error
Create note	Note stored successfully
Generate quiz	Quiz returned
Submit quiz	Score calculated
Unauthorized request	Access denied
Invalid ID	Resource not found

4.3 HTTP Status Codes

The API will use standard HTTP status codes.

200 – Successful request
201 – Resource created
400 – Bad request
401 – Unauthorized
403 – Forbidden
404 – Resource not found
429 – Too many requests
500 – Internal server error

4.4 Project Monitoring

Project progress will be monitored throughout development.

The team will track:

- Completed API endpoints.
- Pending development tasks.
- Testing results.
- Database issues.
- API response time.
- Security issues.
- AI integration problems.
- Documentation completion.

4.5 Planning Conclusion

The Project Planning Phase establishes a clear roadmap for developing the AI StudyBuddy API. It defines the API modules, requirements, resources, technology stack, development schedule, security requirements, testing strategy, and possible risks.

A proper plan ensures that the API is developed systematically and can support the frontend application and AI services effectively.

The planned development flow is:

```
Requirements
  ↓
Project Planning
  ↓
API Design
  ↓
Database Design
  ↓
API Development
  ↓
AI Integration
  ↓
Testing
  ↓
Deployment
  ↓
Maintenance
```

The planning phase therefore provides the foundation for creating a **secure, scalable, reliable, and student-friendly AI StudyBuddy API**.